Geometric Action of Qubit Channels on the Bloch Sphere

1 Bloch Sphere Representation

Any qubit state can be written as

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}), \tag{1}$$

where $\vec{r} = (r_x, r_y, r_z)$ is the Bloch vector with $|\vec{r}| \leq 1$.

- $|\vec{r}| = 1$: pure state
- $|\vec{r}| < 1$: mixed state

—

2 1. Reset Channel

2.1 Kraus Operators

The reset channel maps all states to $|0\rangle$:

$$\mathcal{N}_{\text{reset}}(\rho) = |0\rangle \langle 0| \,. \tag{2}$$

Kraus operators:

$$K_0 = |0\rangle \langle 0| \,, \quad K_1 = |0\rangle \langle 1| \,. \tag{3}$$

They satisfy:

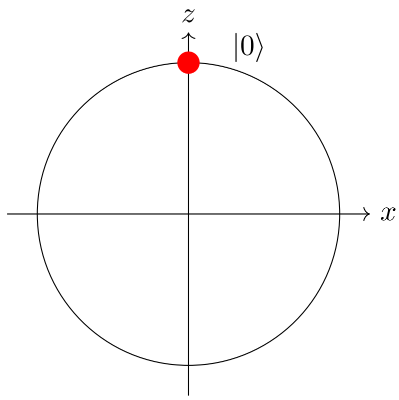
$$K_0^\dagger K_0 + K_1^\dagger K_1 = I.$$

2.2 Bloch Sphere Effect

$$\vec{r} \to (0, 0, 1)$$

- Entire Bloch sphere collapses to a point
- Maximum loss of information

2.3 Diagram



—

3 2. Dephasing Channel

3.1 Kraus Operators

$$K_0 = \sqrt{1-p}I, \quad K_1 = \sqrt{p}Z \tag{4}$$

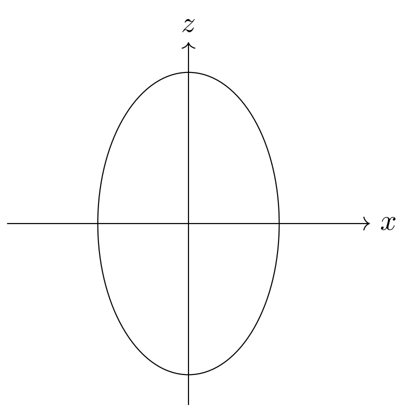
3.2 Bloch Mapping

$$r_x \to (1-2p)r_x, \tag{5}$$
$$r_y \to (1-2p)r_y, \tag{6}$$
$$r_z \to r_z. \tag{7}$$

3.3 Geometric Effect

- Compression along x and y
- z-axis unchanged
- Sphere becomes an ellipsoid

3.4 Diagram



—

4 3. Depolarizing Channel

4.1 Kraus Operators

$$\mathcal{N}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \tag{8}$$

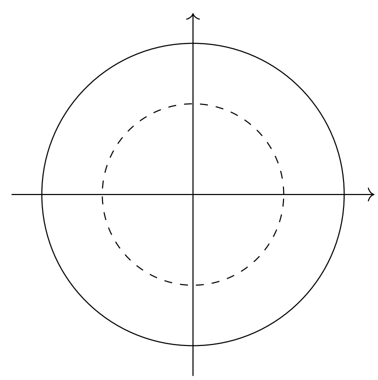
4.2 Bloch Mapping

$$\vec{r} \to (1-p)\vec{r} \tag{9}$$

4.3 Geometric Effect

- Uniform shrinking of the Bloch sphere
- Symmetric noise in all directions
- At $p = 1$: maximally mixed state

4.4 Diagram



—

5 Exam-Style Questions

Question 1

Show that the depolarizing channel maps all pure states toward the maximally mixed state.

Solution:

$$\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma}) \Rightarrow \mathcal{N}(\rho) = \frac{1}{2}(I + (1-p)\vec{r}\cdot\vec{\sigma})$$

Hence $|\vec{r}|$ decreases by factor $(1-p)$.

—

Question 2

Which channel preserves populations but destroys coherence?

Solution:

The dephasing channel:

$$r_x, r_y \to 0, \quad r_z \text{ unchanged}$$

Thus diagonal elements remain unchanged.

—

Question 3

Which channel is non-invertible?

Solution:

The reset channel, since all input states map to the same output state.

—

6 Summary Table

Channel	Effect	Geometry	Invertible
Reset	Full erasure	Point	No
Dephasing	Phase loss	Ellipsoid	Yes (if $p < 1$)
Depolarizing	Uniform noise	Shrinking sphere	No

7 Conclusion

The Bloch sphere provides a powerful geometric picture:

- Reset destroys all information
- Dephasing removes coherence only
- Depolarization uniformly degrades states

These models form the foundation of quantum noise theory.